

$$\begin{pmatrix}
 0 & -a_{21} & \cdots & \cdots & -a_{n1} \\
 a_{21} & \ddots & & & \\
 \vdots & \ddots & \ddots & \ddots & \\
 a_{n1} & \cdots & \cdots & a_{n(n-1)} & 0
 \end{pmatrix}
 \rightarrow
 \begin{pmatrix}
 a_{21} \\
 a_{31} \\
 a_{32} \\
 a_{41} \\
 a_{42} \\
 a_{43} \\
 a_{51} \\
 \vdots \\
 a_{54} \\
 \vdots \\
 a_{n1} \\
 \vdots \\
 a_{n(n-1)}
 \end{pmatrix}$$